

Disproof of conjecture 00000007792: the M-ellipsoid constant of the Euclidean ball is 4, not $\leq (\pi/4)e$

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1 The conjecture and the clause refuted

Statement (English, verbatim). Definition: The M-ellipsoid condition: an M-position of a convex body K is an ellipsoid E with $\text{vol}(K + E) \text{vol}(K^\circ + E^\circ) \leq C^n \text{vol}(K) \text{vol}(K^\circ)$. Conjecture: The optimal constant C is uniformly attainable over all bodies with $C \leq (\pi/4)e$, given asymptotically by the John ellipsoid position; and M-positions are stable: bodies deviating from the optimal position lose exponentially and strictly monotonically in volume product, with characterization function e^{cd^2} in the position deviation d . (The Chinese text states the same: the optimal constant C is uniformly attainable over all bodies and $C \leq \pi/4 \cdot e$.)

Reading. The conjecture is a conjunction. Its first conjunct says that a single constant $C \leq (\pi/4)e \approx 2.135$ works for all convex bodies K (in every dimension n), i.e. every K has an ellipsoid E with

$$\text{vol}(K + E) \text{vol}(K^\circ + E^\circ) \leq C^n \text{vol}(K) \text{vol}(K^\circ),$$

the inequality exactly as written in the Definition line (with C^n , not C^{2n}). We refute this conjunct, which refutes the conjunction. The stability conjunct (the e^{cd^2} law) is not addressed. We refute three readings of the first conjunct:

- (R1) one $C \leq (\pi/4)e$ is admissible in every dimension $n \geq 1$;
- (R2) (“asymptotically”) one $C \leq (\pi/4)e$ is admissible in all sufficiently large dimensions;
- (R3) admissible constants $C_n \geq 0$ in dimension n (for all large n) converge to a limit $L \leq (\pi/4)e$.

In fact, in every dimension $n \geq 1$ every admissible constant satisfies $|C| \geq 4$. A reading with $C \leq \pi/(4e)$ is stronger than (R1) and is refuted too. The witness is the Euclidean unit ball, and the bound 4 holds against *every* ellipsoid, so no choice of position (John or otherwise) helps.

Conventions. Space $\mathbb{R}^n$ with the Euclidean inner product (Mathlib `EuclideanSpace R (Fin n)`); vol is Lebesgue measure (Mathlib `volume`, valued in $[0, \infty]$); $K + E$ is the Minkowski sum. Polarity is with respect to the origin. Ellipsoids may have any centre. The constant C is real; we prove $|C| \geq 4$ in each dimension, and $C \geq 4$ in odd dimensions (where $C^n < 0$ is excluded automatically).

2 Definitions (as formalized)

Definition 1. $B = \{x : \|x\| \leq 1\}$. The *polar* of $K \subseteq \mathbb{R}^n$ is $K^\circ = \{y : \langle x, y \rangle \leq 1 \text{ for all } x \in K\}$ (`polar`). An *ellipsoid* is a set $E = \{c + Ax : x \in B\}$ with $c \in \mathbb{R}^n$ and A an invertible linear map (`IsEllipsoid`). A *convex body* is a compact convex set with nonempty interior (`IsConvexBody`).

Definition 2. E satisfies the M -ellipsoid condition for K with constant C (`MEllipsoidCondition`) if E is an ellipsoid and $\text{vol}(K + E) \text{vol}(K^\circ + E^\circ) \leq C^n \text{vol}(K) \text{vol}(K^\circ)$, where C^n enters through $\max(C^n, 0)$ (`ENNRReal.ofReal`). C is admissible in dimension n (`AdmissibleConstant n C`) if every convex body $K \subseteq \mathbb{R}^n$ with $0 \in \text{int } K$ has an ellipsoid E satisfying the condition.

3 Proof

Lemma 3 (`polar_closedBall`, `closedBall_isConvexBody`). $B^\circ = B$, and B is a convex body with 0 in its interior.

Proof. If $\|y\| \leq 1$ then $\langle x, y \rangle \leq \|x\| \|y\| \leq 1$ on B . If $y \in B^\circ$, $y \neq 0$, test $x = y/\|y\|$ to get $\|y\| \leq 1$. $\square$

Lemma 4 (`key_ineq`). For $d, x \in \mathbb{R}^n$ with $\|x\| \leq 1$ and $k = (1 + \|d\|)^{-1}$: $\langle d, x - kd \rangle + \|x - kd\| \leq 1$.

Proof. Put $r = \|d\|$, $p = \langle d, x \rangle \leq r$. Then $\langle d, x - kd \rangle = p - kr^2$, and $R := 1 - p + kr^2 \geq 1 - r + kr^2 = k > 0$. Using $k(1 + r) = 1$ one checks the identity $R^2 - \|x - kd\|^2 = (r - p)^2 + 1 - \|x\|^2 \geq 0$, so $\|x - kd\| \leq R$. $\square$

Theorem 5 (`ball_volume_product_ge`). For every n and every ellipsoid $E \subseteq \mathbb{R}^n$: $4^n \text{vol}(B) \text{vol}(B^\circ) \leq \text{vol}(B + E) \text{vol}(B^\circ + E^\circ)$.

Proof. Let $E = c + AB$. The map A^*A is symmetric; let $v_1, \dots, v_n$ be an orthonormal eigenbasis, $A^*Av_j = \lambda_j v_j$ (`Mathlib LinearMap.IsSymmetric.eigenvectorBasis`). Then $\langle Av_i, Av_j \rangle = \lambda_j \delta_{ij}$, so $\lambda_j = \|Av_j\|^2 > 0$. Put $s_j = \sqrt{\lambda_j}$ and $u_j = s_j^{-1} Av_j$; the u_j form an orthonormal basis. Let R be the isometry with $Ru_j = v_j$, and for $a \in \mathbb{R}^n$ let D_a be the linear map with $D_a u_j = a_j u_j$, so $\det D_a = \prod_j a_j$ and $D_{1+a} = I + D_a$. Checking on the basis u_j : $AR = D_s$ and $A^*D_{s^{-1}} = R$.

First factor. For $x \in B$, $c + D_{1+s}x = x + (c + A(Rx)) \in B + E$ because $\|Rx\| = \|x\|$. Hence $B + E \supseteq c + D_{1+s}B$ and, by translation invariance and the linear change of variables (`addHaar_image_linearMap`), $\text{vol}(B + E) \geq \prod_j (1 + s_j) \text{vol}(B)$.

Second factor. Let $d = D_{s^{-1}}^* c$ and $k = (1 + \|d\|)^{-1}$. For $x, x' \in B$ and $z = x - kd$:

$$\langle c + Ax', D_{s^{-1}}z \rangle = \langle d, z \rangle + \langle x', A^*D_{s^{-1}}z \rangle = \langle d, z \rangle + \langle x', Rz \rangle \leq \langle d, z \rangle + \|z\| \leq 1$$

by the key inequality. So $D_{s^{-1}}(x - kd) \in E^\circ$, and $x + D_{s^{-1}}(x - kd) = -kD_{s^{-1}}d + D_{1+s^{-1}}x \in B + E^\circ$. Hence $\text{vol}(B^\circ + E^\circ) = \text{vol}(B + E^\circ) \geq \prod_j (1 + s_j^{-1}) \text{vol}(B)$.

Product. $(1 + s)(1 + s^{-1}) - 4 = (s - 1)^2/s \geq 0$ for $s > 0$ (`four_le_mul_inv`), so the product of the two lower bounds is at least $4^n \text{vol}(B)^2 = 4^n \text{vol}(B) \text{vol}(B^\circ)$. $\square$

Corollary 6 (`ball_condition_pow_ge`, `admissible_abs_ge_four`, `admissible_ge_four_of_odd`). If some ellipsoid satisfies the M -ellipsoid condition for B with constant C , then $4^n \leq C^n$. Hence in every dimension $n \geq 1$ every admissible C has $|C| \geq 4$, and $C \geq 4$ when n is odd.

Proof. Combine the theorem with the hypothesis and cancel $\text{vol}(B)^2 \in (0, \infty)$; if $C^n \leq 0$ the right side is 0, contradicting $4^n > 0$. $\square$

Corollary 7 (`uniform_admissible_ge_four`, `conjecture_7792_false`, `conjecture_7792_false_eventually`, `conjecture_7792_false_limit`). Since $(\pi/4)e < e < 3 < 4$: (R1) fails (a uniformly admissible C is ≥ 4 , using $n = 1$); (R2) fails (use the odd dimension $2N + 1$); (R3) fails (eventually $C_n \geq 4$, so $L \geq 4$).

The bound is sharp for the ball: $E = B$ gives $\text{vol}(2B)^2 = 4^n \text{vol}(B)^2$ (not formalized; not needed).

4 Lean formalization and axiom audit

File `lean/Conjecture7792/Basic.lean` (332 lines, import `Mathlib` only, Lean v4.33.1, `Mathlib` v4.33.1). Main theorem:

```
theorem conjecture_7792_false :  
  Not (exists C : Real, C <= Real.pi / 4 * Real.exp 1 /\  
    forall n : Nat, 1 <= n -> AdmissibleConstant n C)
```

(ASCII rendering of the Lean statement) together with `conjecture_7792_false_eventually`, `conjecture_7792_false_limit`, `admissible_abs_ge_four` and the core estimate `ball_volume_product_ge`. `lake build` succeeds. `#print` axioms for each of these reports only `propext`, `Classical.choice`, `Quot.sound`. There is no `sorry` and no `native_decide`.